Objects and Classes - Questions & Answers

1 Mark Questions

1. What is a class in Java?

   - A class is a blueprint or template from which objects are created.

2. What is an object?

   - An object is an instance of a class.

3. What keyword is used to create an object in Java?

   - The 'new' keyword.

4. Name two types of Java constructors.

   - Default constructor and parameterized constructor.

5. What is an anonymous object?

   - An object without a reference name, created and used only once.

2 Mark Questions

1. How are objects initialized in Java?

   - Objects can be initialized using:

     - Reference variable

     - Method

     - Constructor

2. What are the rules for creating a Java constructor?

   - The constructor must have the same name as the class and cannot have a return type.

3. Differentiate between instance and static variables.

   - Instance variables belong to an object and are created at runtime, while static variables belong to the class and are shared by all objects.

4. Why do we use the 'this' keyword in Java?

- The 'this' keyword refers to the current object and is used to avoid naming conflicts and call methods or constructors.

3 Mark Questions

1. What is constructor overloading? Explain with an example.

- Constructor overloading is when a class has multiple constructors with different parameter lists.
- Example:

```
class Box {
    double width, height, depth;
    Box(double w, double h, double d) { width = w; height = h; depth = d; }
    Box() { width = height = depth = 0; }
}
```

2. Explain method overloading with an example.

- Method overloading occurs when multiple methods have the same name but different parameters.
- Example:

```
class MathOperations {
    void sum(int a, int b) { System.out.println(a + b); }
    void sum(double a, double b) { System.out.println(a + b); }
}
```

3. What is a static method? How is it different from an instance method?

- A static method belongs to a class and can be called without creating an object, while an instance method belongs to an object and requires instantiation.

5 Mark Questions

1. Compare a constructor and a method in Java.

- Constructor:

- Used to initialize an object.
- Has the same name as the class.
- No return type.

- Method:
- Used to define behavior.
- Can have any name.
- Has a return type.

2. Explain the use of the 'static' keyword in Java with examples.

- Static variable: Shared among all objects of a class.
- Static method: Called without an object.
- Static block: Runs when the class is loaded.
- Example:

```
class Example {
    static int count = 0;
    static void show() { System.out.println("Static method"); }
    static { System.out.println("Static block executed"); }
}
```